# Navroop Singh


Systems Engineer

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| SHEET | TITLE | ISSUED | STATUS |
| --- | --- | --- | --- |
| R.01 | Résumé | Aug 2026 | Open to roles |

## PROFILE

Systems engineer with a mechanical engineering foundation and two years of industrial commissioning experience in a polysilicon process plant, now an M.Sc. Systems Engineering student at Otto von Guericke University Magdeburg. Work spans rotating equipment, maintenance engineering and P&ID / GAD interpretation through to equipment installation, commissioning and plant optimization. Moving toward MBSE, system architecture and simulation-driven production systems.

**Availability:** up to 20 h/week during semester, full-time in breaks · **Earliest start:** immediate · **Location:** Magdeburg, open to relocating anywhere in Germany and to remote work · **Work authorisation:** non-EU student, eligible for working-student employment under German student visa terms.

## EXPERIENCE

Aug 2024 — Sep 2025
Jamnagar, India

### Rotating Equipment Engineer — Installation & Commissioning

Reliance Corp IT Park Ltd (Reliance Industries — New Energy) — Polysilicon Plant · designation: Manager

- Supervised installation of three reciprocating compressors — two horizontal, one vertical — covering vendor coordination, site sequencing and quality inspection through to pre-commissioning.
- Witnessed the Mechanical Run Test at the vendor's works for one compressor, and one canned seal-less pump tag on site.
- Reviewed and interpreted General Arrangement Drawings (GAD) and P&IDs for rotating equipment installations.
- Supervised nitrogen preservation procedures ahead of commissioning.
- Developed a reactor bell shifting and sequencing plan projected to remove ~14 h of downtime per batch (modelled — plant in construction phase).

Aug 2023 — Aug 2024
Jamnagar, India

### Graduate Engineering Trainee

Reliance Corp IT Park Ltd (Reliance Industries — New Energy) — Polysilicon Plant

- Assisted the rotating equipment team during installation, assembly and pre-commissioning.
- Supported coordination between maintenance teams, vendors and project engineers.
- Participated in documentation reviews, quality inspections and MRT observations.

Dec 2021 — Jun 2022
Faridabad, India

### R&D Intern

Escorts Kubota Limited — Agricultural & Construction Machinery

- Built an Excel/VBA kinematics and force-analysis template for backhoe loader linkages, automating geometry and free-body calculations previously done by hand.
- Cut manual calculation time for linkage kinematic analysis by ~30%.
- Modelled a quick-hitch coupler in Siemens Solid Edge.
- Internship formally certified by Escorts Kubota HR (EAM & ECE – R&D).

## EDUCATION

2025 — Expected 2027
Magdeburg, DE

### M.Sc. Systems Engineering for Manufacturing

Otto von Guericke University Magdeburg

Systems Engineering · MBSE · System Architecture · Systems Modeling (SysML, UML) · Control Systems · Systems Integration · Modelling & Simulation of Logistics Planning · Python in Production Systems · Engineering of Resilient Production Systems · Human-Technology Interaction

2019 – 2023
Patiala, IN

**B.Eng. Mechanical Engineering**
Thapar Institute of Engineering and Technology — German scale 2.1

Machine Design · Mechanical Vibrations · Heat Transfer · Fluid Mechanics · Manufacturing Processes · Strength of Materials · Dynamics & Vibrations

## SELECTED PROJECTS

### Backhoe Loader Kinematics & Force Analysis `ESCORTS KUBOTA`

Excel/VBA kinematics and force-analysis template for backhoe loader linkages — linkage geometry driving a full free-body solution at every pin joint, with computed reach, working height, bucket rotation and breakout force. Cut manual calculation time by ~30%.

### PPR Modelling & Validation Tool `GROUP OF TWO`

Interactive web app modelling Product-Process-Resource relationships as a directed graph — import from JSON, XML or Excel, in-place editing with PPR conformance enforced at insertion time, two engineering views (basic engineering and reliability) over the shared graph, and four requirement-check algorithms. Deployed on Streamlit Cloud. Group of two; my scope: the engineering and reliability views. `Python · Streamlit · NetworkX · Graphviz · PyVis`

### Production Scheduling & Lot-Sizing Study `SUMMER SEMESTER 2026`

Modelled a six-product, five-stage line (Press → Laser → Bending → Profiling → Welding, two-machine Profiling bottleneck) in Tecnomatix Plant Simulation and benchmarked five order-release strategies over 100 orders / 1,050 units against makespan, total tardiness and on-time delivery. A lot-based search cut makespan 20% and tardiness 64% against the family-production baseline, but EDD eliminated tardiness entirely and won the weighted objective — so the recommendation is lot-based release now, EDD after a SMED set-up-reduction programme. `Tecnomatix Plant Simulation 2404 · SimTalk`

### Thermal Printer MBSE Model (RAMI 4.0) `GROUP OF TWO`

Modelled a thermal printer in Enterprise Architect against the RAMI 4.0 reference architecture — two hierarchy levels (Work Station, Control Device), the six RAMI layers plus a Product-Process-Resource layer, refined top-down with traceability from workstation goals to physical realisation. My scope: the Sheet Feed Controller in full — all seven layers, every diagram and its traceability to the workstation requirements — plus the Business, Function and Information layers at Work Station level.

### Medical Exoskeleton `TEAM LEAD`

Led a five-member capstone team designing a low-cost lower-limb exoskeleton — sized to a 78 kg user and 150 Nm swing torque, load calculations and component sizing in aluminium 6061 (FOS 4) with 304 stainless steel shafts, eight parts checked in FEA. Delivered a working prototype at €300, fully reimbursed by the institute.

## SKILLS

| | |
|---|---|
| SYSTEMS ENGINEERING | MBSE · SysML · UML · System Architecture · RAMI 4.0 |
| CAD & DESIGN | SolidWorks · PTC Creo · AutoCAD · Siemens Solid Edge · Navisworks |
| MODELLING & SIMULATION | ANSYS · Siemens Tecnomatix Plant Simulation · Enterprise Architect (Sparx) |
| PROGRAMMING | Python · MATLAB · SimTalk · Streamlit · NetworkX · Graphviz |
| PLANT & DOMAIN | Rotating equipment · Maintenance engineering · P&ID / GAD · Installation & commissioning · Plant optimization |
| STANDARDS | ISO 13482 · ISO 13485 · ISO 14001 (working familiarity) |
| LANGUAGES | English (C1) · German (A2) · Punjabi (native) · Hindi (fluent) |

## COMMUNITY

2025 – Present
Magdeburg, DE

**Volunteer Member — DEGIS**
Otto von Guericke University Magdeburg

Planning and running cultural events, and coordinating society activities across a multicultural team.

2025 – Present
Magdeburg, DE

**Member — ENACTUS**
Student entrepreneurship network

Attended the ENACTUS training weekend in Heidelberg in 2025 — workshops on time management and on the startup accelerator programme.